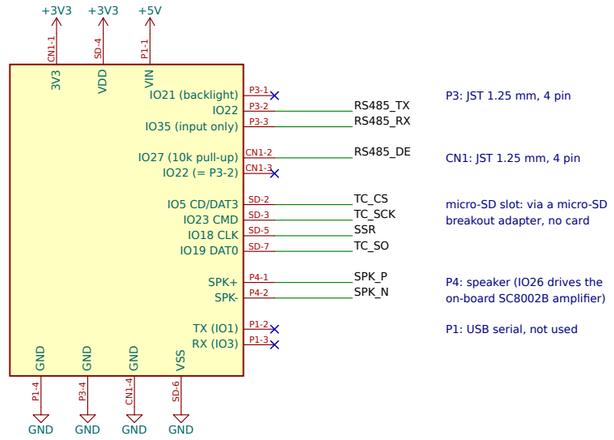
CYD ESP32-2432S028R

ESP32-2432S028R (CYD)
U1



P3: JST 1.25 mm, 4 pin

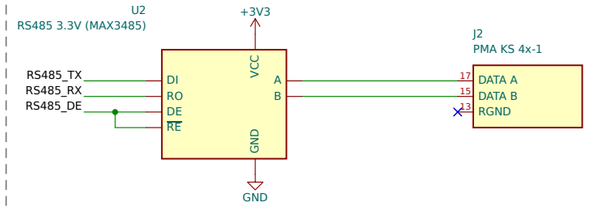
CN1: JST 1.25 mm, 4 pin

micro-SD slot: via a micro-SD breakout adapter, no card

P4: speaker (IO26 drives the on-board SC8002B amplifier)

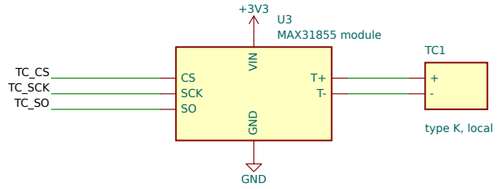
P1: USB serial, not used

RS485 (Modbus RTU, beta)



One or two controllers on the same A / B pair (own addresses).
No answer: swap A / B. No 120 ohm terminators below 8 devices.
RGND not connected. Auto-direction module: DE / RE open.

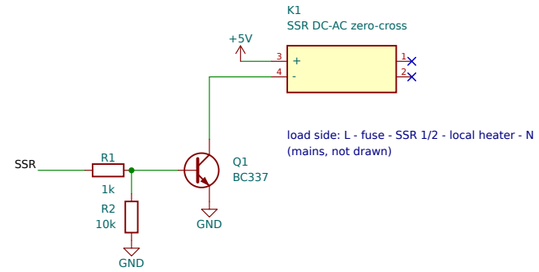
Thermocouples (local channel)



type K, local

The local channel: bottom or top (setting), the other one Modbus.
MAX6675 modules fit the same way.

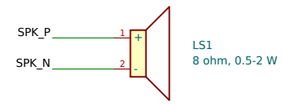
SSR driver and SSR (local channel)



load side: L - fuse - SSR 1/2 - local heater - N
(mains, not drawn)

Q: BC337 or 2N2222 (TO-92 C-B-E). The 10k base pull-down keeps the SSR off during reset and flashing.
SSR + input: 5 V from P1-1 (VIN).

Alarm sound (speaker on P4)



IO26 drives the CYD's SC8002B amplifier, its bridged output is P4: SPK- is not ground, connect the speaker only between SPK+ and SPK-. A small 8 ohm speaker or a passive piezo. 2.7 kHz beeps (setting safety.sound; touch: Settings > System > Alarm sound).

Notes

TempCtl Duo, env cyd-1xlocal-1x485; pins as in src/board_cyd.h (firmware 1.0.39). Modules connected with cables: a connection schematic of the whole device, not a PCB layout.
Net labels with the same name are connected.

Power: USB on the CYD. 3V3 for the modules from CN1-1 or the SD VDD pin (the board's regulator), 5 V for the SSR inputs from P1-1 (VIN). One ground for the CYD and every module.

The SD-slot pins: plug a micro-SD breakout (sniffer) adapter into the slot; no SD card. IO5 has the board's SD pull-up (converter deselected at boot); IO18 has none (SSR off with the 10k pull-down).

No E-stop input, no SSR supply relay and no chip temperature on the CYD (no free pins). IO35 is input only.

A software controller is never the only protection: fit an independent over-temperature limiter and a contactor.



one ground for the CYD and every module

Modules connected with cables; pins: src/board_cyd.h

Title: **TempCtl Duo - CYD A: RS485 + one local channel**

env cyd-1xlocal-1x485, firmware 1.0.39

Size: A3 | Date: 2026-10-13 | Rev: 1.0.39 | Sheet: 1/1

File: TempCtl-Duo-CYD-A.kicad_sch